

DSA Day 10: Sorted & Sorting a Linked List

Book: Data Structures Through C in Depth - Ch3 §3.5-3.6 · pp 83-98 | Code in Python

Overview

Keep a list sorted on insert, or sort an existing one. You can sort by swapping data or by rearranging links.

Key concepts to master

- Sorted insert
- Selection/bubble by data
- Selection/bubble by relinking
- Data-swap vs link-rearrange trade-off

Python implementation

```
def sorted_insert(head, v):
    n=Node(v)
    if not head or v<head.data: n.next=head; return n
    cur=head
    while cur.next and cur.next.data<v: cur=cur.next
    n.next=cur.next; cur.next=n; return head
```

Complexity

Sorted insert $O(n)$; bubble/selection $O(n^2)$.

Common pitfalls

- Relinking bugs when nodes are adjacent
- Not handling insert before head

Practice

- Insert into a sorted list
- Bubble-sort by relinking

